CHAPTER 2 LIMITS AND DERIVATIVES

2.2 The Limit of a Function

- Calculus is the study of limits.

- How do we find the limit of $x^2 - x + 2$ as x approaches 2?

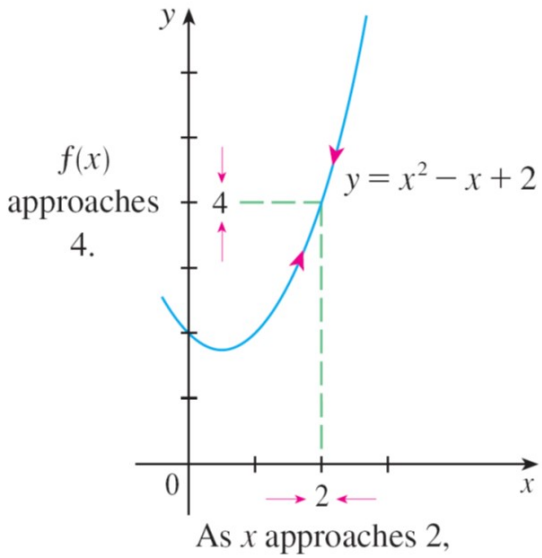
not equal to

x	$f(x)$	x	$f(x)$
1.0	2.000000	3.0	8.000000
1.5	2.750000	2.5	5.750000
1.8	3.440000	2.2	4.640000
1.9	3.710000	2.1	4.310000
1.95	3.852500	2.05	4.152500
1.99	3.970100	2.01	4.030100
1.995	3.985025	2.005	4.015025
1.999	3.997001	2.001	4.003001

numerical method

increase to 2

decrease



As x approaches 2,

$$A^3 - B^3 = (A-B)(A^2 + AB + B^2)$$

- **Definition**

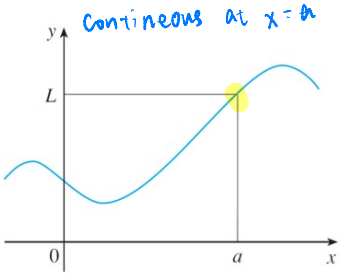
Suppose that $f(x)$ is defined when x is *near* the number a. We write

$$\lim_{x \to a} f(x) = L$$

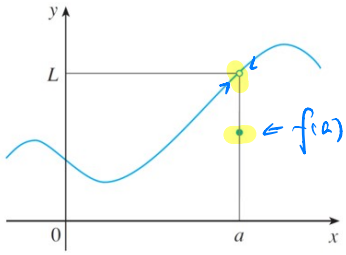
and say that

the **limit** of $f(x)$, as x approaches a, equals L

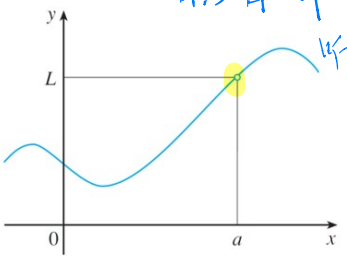
if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a (on either side of a) but *not equal to* a.



(a)



(b)



(c)

Example 2.1. Find $\lim_{x \to 3} (4x - 5)$.

x	$4x - 5$
2.9	6.600
2.99	6.960
2.999	6.996
3.001	7.004
3.01	7.040
3.1	7.400

Example 2.2. Find $\lim_{x\to 3} \dfrac{x^2-x-6}{x-3}$.

x	$\dfrac{x^2-x-6}{x-3}$
2.9	4.900
2.99	4.990
2.999	4.999
3.001	5.001
3.01	5.010
3.1	5.100

Example 2.3. Find $\lim_{x\to 0} \dfrac{\sqrt{x^2+9}-3}{x^2}$.

x	$\dfrac{\sqrt{x^2+9}-3}{x^2}$
± 0.01	0.166666203704757
± 0.001	0.166666661804982
± 0.0001	0.166666680456728
± 0.00001	0.166666680456728

若分子有带根分数 => 先化简到非分数形式

=> (√ + _)(√ - _)

eg $\lim_{x\to 4} \dfrac{\left(\frac{1}{\sqrt{x}} - \frac{1}{2}\right) 2\sqrt{x}}{(x-4)\,2\sqrt{x}}$

$= \lim_{x\to 4} \dfrac{2 - \sqrt{x}\,(2+\sqrt{x})}{(x-4)\,2\sqrt{x}\,(2+\sqrt{x})}$

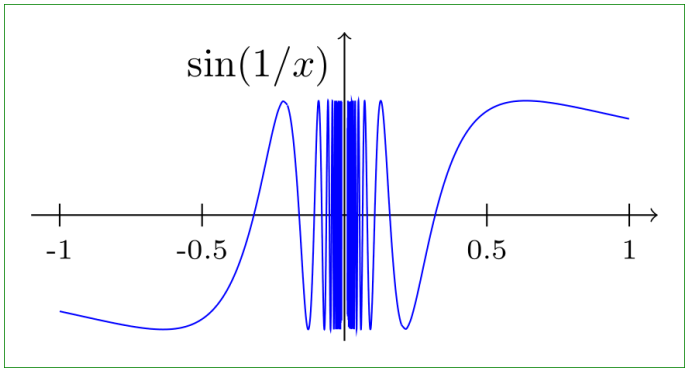
$= \lim_{x\to 4} \dfrac{4 - x}{2\sqrt{x}\,(x-4)(2+\sqrt{x})}$

$= -\lim_{x\to 4} \dfrac{1}{2\sqrt{x}(2+\sqrt{x})}$

$= -\lim_{x\to 4} \dfrac{1}{4\sqrt{4}(2+\sqrt{4})}$

$= -\dfrac{1}{32}$

Example 2.4. Find $\lim_{x\to 0} \sin\left(\frac{1}{x}\right)$. DNE (doesn't Exist)



x	$\frac{\sqrt{x^2+9}-3}{x^2}$
± 0.01	-0.506365641109759
± 0.001	0.826879540532003
± 0.0001	-0.305614388888252
± 0.00001	0.035748797986561
± 0.000001	-0.349993502171308
± 0.0000001	0.420547793190927

- **Definition**

We write

$$\lim_{x\to a^-} f(x) = L$$

↓ $a^- < a$

and say that

the **left-hand** limit of $f(x)$, as x approaches a from the left, equals L if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a with x less than a.

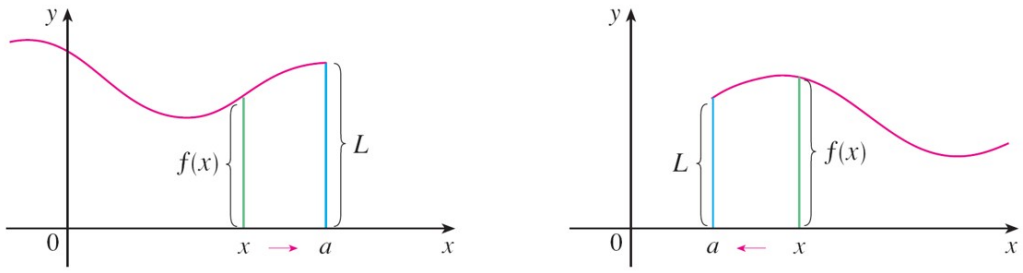
- **Definition**

We write

$$\lim_{x \to a^+} f(x) = L$$

and say that

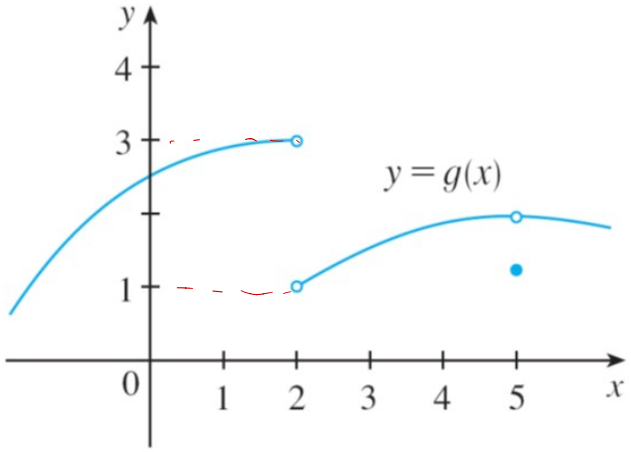
the **right-hand** limit of $f(x)$, as x approaches a from the right, equals L if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a with x greater than a.



- **Theorem** 判断极限是否存在.

$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^+} f(x) = L$ and $\lim_{x \to a^-} f(x) = L$.

Example 2.5. The graph of a function g is shown below. Use it to state the values (if they exist) of the following:



(a) $\lim_{x\to 2^-} g(x)$ DNE $\neq 3$ x=2 有没有见义不重要

(b) $\lim_{x\to 2^+} g(x)$ $= 1$

(c) $\lim_{x\to 2} g(x)$ DNE since $\lim_{x\to 2^-} g(x) \neq \lim_{x\to 2^+} g(x)$

(d) $\lim_{x\to 5^-} g(x)$ $= 2$

(e) $\lim_{x\to 5^+} g(x)$ $= 2$

(f) $\lim_{x\to 5} g(x)$ $= 2$ since $\lim_{x\to 5^-} g(x) = \lim_{x\to 5^+} g(x)$ (答案要有理由)

(g) $g(5)$ $= 1.5$

- The Heaviside function H is define by

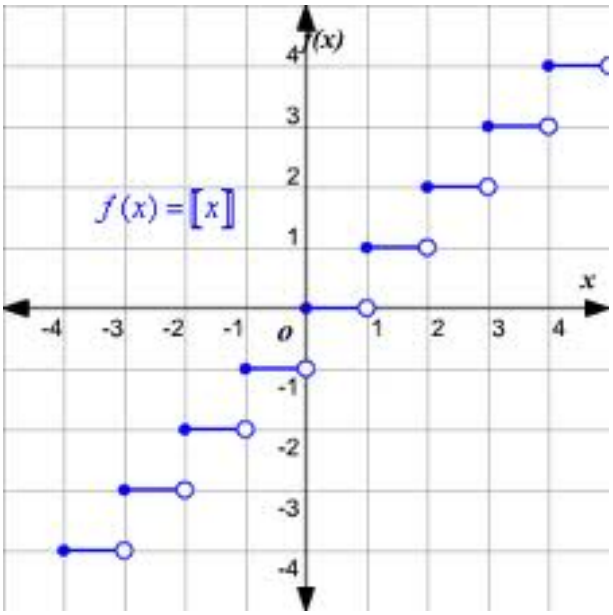
$$H(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$

- Find $\lim_{t \to 0^-} H(t)$ and $\lim_{t \to 0^+} H(t)$.

$\lim_{t \to 0^-} H(t) = 0$

$\lim_{t \to 0^+} H(t) = 1$

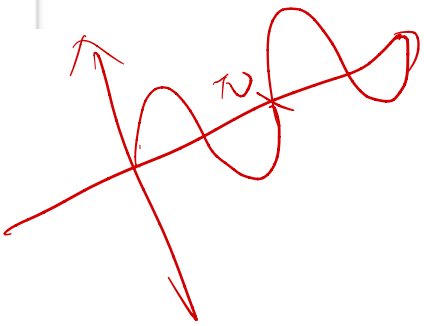
- $[\![x]\!]$ denotes the greatest integer less than or equal to x.



$[\![2]\!] = 2$

$[\![2.1]\!] = 2$

$[\![2.9]\!] = 2$



Example 2.6. Find $\lim_{x \to 2} [\![x]\!]$.

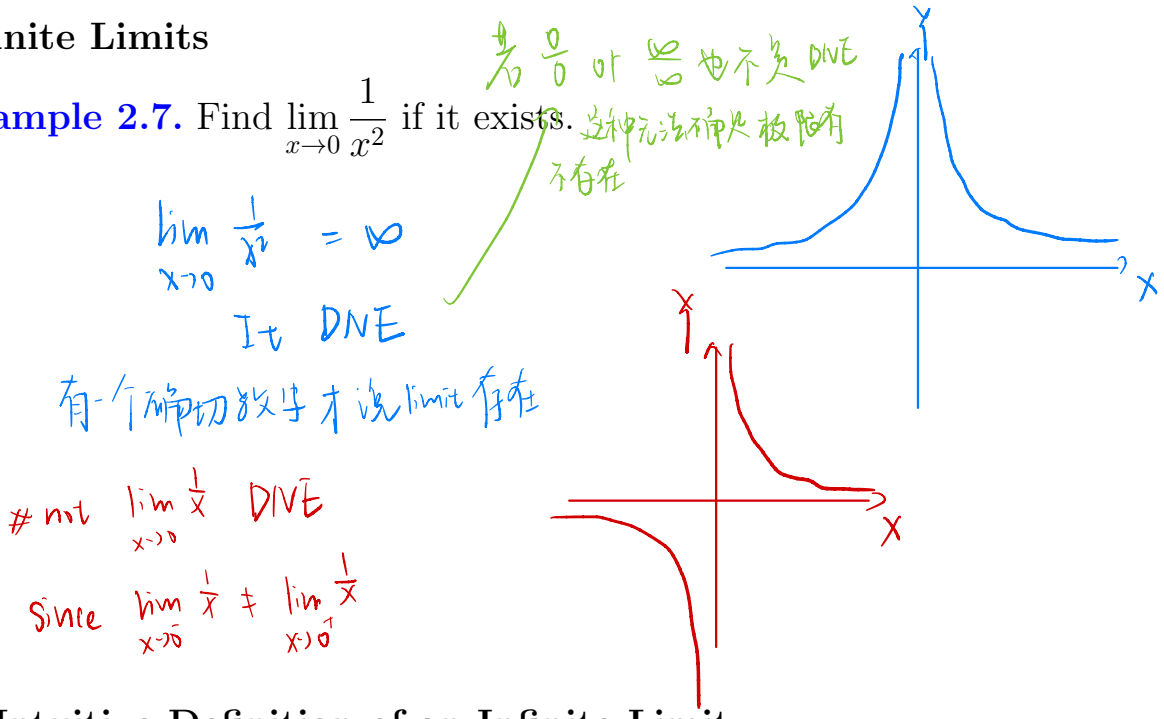
$\lim_{x \to 2^-} [\![x]\!] = 1$

$\lim_{x \to 2^+} [\![x]\!] = 2$

$\lim_{x \to 2} [\![x]\!]$ DNE, since $\lim_{x \to 2^-} [\![x]\!] \neq \lim_{x \to 2^+} [\![x]\!]$

Infinite Limits

Example 2.7. Find $\lim_{x\to 0} \frac{1}{x^2}$ if it exists.



- **Intuitive Definition of an Infinite Limit**

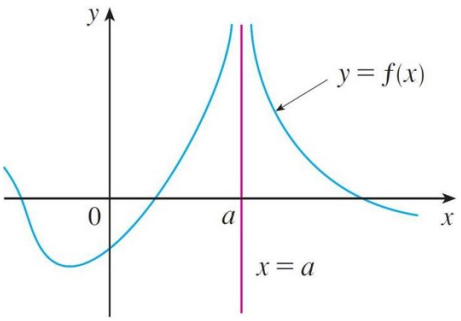
Let f be a function defined on both sides of a, except possibly at a itself. Then

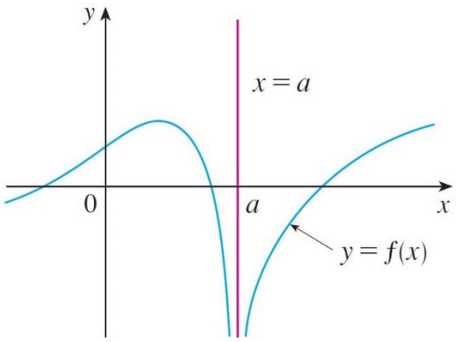
$$\lim_{x\to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a, but not equal to a.

$$\lim_{x\to a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.





Example 2.8. Find $\lim_{x\to 3^+} \dfrac{1}{x-3}$ and $\lim_{x\to 3^-} \dfrac{1}{x-3}$ if they exist.

$= \infty$ $= -\infty$

Example 2.9. Find $\lim_{x\to -3^+} \dfrac{2x}{x+3}$ and $\lim_{x\to -3^-} \dfrac{2x}{x+3}$.

$= \lim_{x\to -3^+} \dfrac{2}{1+\frac{3}{x}}$ $= +\infty$

不可以代入

$= -\infty$

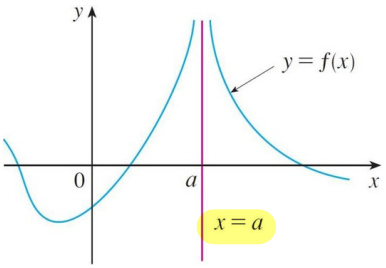
$\dfrac{-6}{0}$

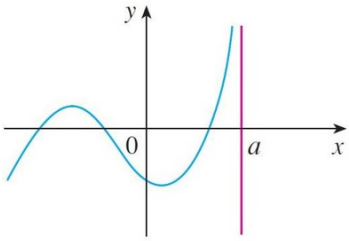
- **Vertical Asymptotes**

The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true.

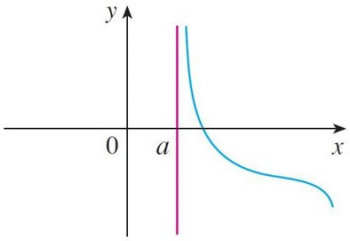
$$\lim_{x\to a} f(x) = \infty \qquad \lim_{x\to a^-} f(x) = \infty \qquad \lim_{x\to a^+} f(x) = \infty$$

$$\lim_{x\to a} f(x) = -\infty \qquad \lim_{x\to a^-} f(x) = -\infty \qquad \lim_{x\to a^+} f(x) = -\infty$$

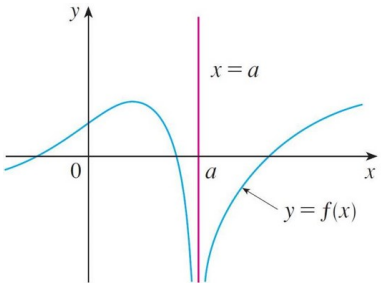


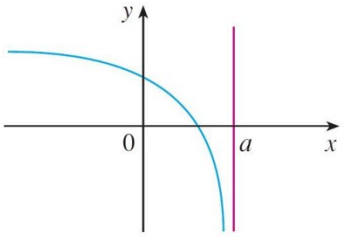


(a) $\lim_{x\to a^-} f(x) = \infty$

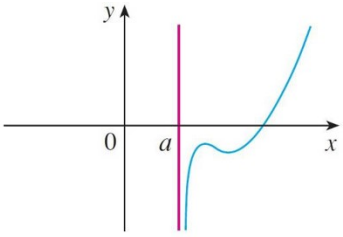


(b) $\lim_{x\to a^+} f(x) = \infty$





(c) $\lim_{x\to a^-} f(x) = -\infty$



(d) $\lim_{x\to a^+} f(x) = -\infty$

Example 2.10. Find the vertical asymptote of $f(x) = \ln x$.

Domain is $x > 0$

$\lim_{x\to 0^+} f(x) = \lim_{x\to 0^+} \ln x = -\infty$

$f(x)$ has a vertical asymptotes at $x = 0$

Example 2.11. Find the vertical asymptotes of $f(x) = \tan x$.

$\tan x = \frac{\sin x}{\cos x}$

$\tan x$ is undefined when $\cos x = 0$, $x = \frac{\pi}{2} + 2n\pi$, $n = 0, 1, 2, \cdots$

or $\frac{3\pi}{2} + 2n\pi$

$x = \frac{\pi}{2} + n\pi$, $n \in \mathbb{Z}$

$\lim_{x \to \frac{\pi}{2}^-} \tan x = \infty$

$\lim_{x \to \frac{\pi}{2}^+} \tan x = \infty$

$\therefore \tan x$ has an asymptotes line at $x = \frac{\pi}{2}$. However, $\tan x$ is a periodic function, thus it has an asymptote line at $x = \frac{\pi}{2} + n\pi$

Exercises 2.2 5, 7, 9, 11, 19, 23, 27, 31, 33, 35, 41, 43, 45(a) + find the vertical asypmtote of the function.

2.3 Calculate Limit Using the Limit Laws

- **Limit Laws**

Let c be a constant, and f and g be functions that have limits at a. Then

1. $\lim_{x\to a} cf(x) = c\lim_{x\to a} f(x)$.
2. $\lim_{x\to a} [f(x) + g(x)] = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$.
3. $\lim_{x\to a} [f(x) - g(x)] = \lim_{x\to a} f(x) - \lim_{x\to a} g(x)$.
4. $\lim_{x\to a} [f(x) \cdot g(x)] = \lim_{x\to a} f(x) \cdot \lim_{x\to a} g(x)$.
5. $\lim_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ provided that $\lim_{x\to a} g(x) \neq 0$.
6. $\lim_{x\to a} c = c$.
7. $\lim_{x\to a} x = a$.
8. $\lim_{x\to a} [f(x)]^n = \left[\lim_{x\to a} f(x)\right]^n$, n is a positive integer.
9. $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$.

不包括 complex number

Example 2.12. Find $\lim_{x\to 3} 2x^4$.

Domain : $x \in \mathbb{R}$ (real number)

$= 2 \lim_{x\to 3} x^4$

$= 162$

Example 2.13. Find $\lim_{x\to 4}\left(3x^2-2x\right)$. D: $x\in\mathbb{R}$

$= \lim_{x\to 4}(3\times 4^2 - 2\times 4)$

$= \lim_{x\to 4} 40$

Example 2.14. Find $\lim_{x\to 4}\dfrac{\sqrt{x^2+9}}{x}$.

$= \lim_{x\to 4}\dfrac{\sqrt{16+9}}{4}$

$= \lim_{x\to 4}\dfrac{5}{4}$

$= \lim_{x\to 4} 1.25$

Example 2.15. If $\lim_{x\to 3} f(x)=4$ and $\lim_{x\to 3} g(x)=8$, find $\lim_{x\to 3}\left[f^2(x)\cdot\sqrt[3]{g(x)}\right]$.

use limit laws

$= \lim_{x\to 3} f^2(x) \cdot \lim_{x\to 3}\sqrt[3]{g(x)}$

$= \left[\lim_{x\to 3} f(x)\right]^2 \times \sqrt[n]{\lim_{x\to 3} g(x)}$

$= 4^2 \times \sqrt[3]{8}$

$= 16\times 2 = 32$

- **Substitution Property**

If f is a **polynomial function** or a **rational function**, then

$$\lim_{x\to a} f(x) = f(a)$$

provided that $f(a)$ is defined. 若 a 在 domain 里面，就可以 direct substitute

Example 2.16. Find $\lim_{x\to 2} \dfrac{7x^5 - 10x^4 - 13x + 6}{3x^2 - 6x - 8}$.

$= \lim_{x\to 2} \dfrac{7\times 2^5 - 10\times 2^4 - 13\times 2 + 6}{3\times 2^2 - 6\times 2 - 8}$

$= -\dfrac{11}{2}$

$(x-1)^2$ always > 0 (可以不用代值)

Example 2.17. Find $\lim_{x\to 1} \dfrac{x^3 + 3x + 7}{x^2 - 2x + 1}$.

limit doesn't exit 的情况 check 是哪种 asymptotes line

$= \dfrac{11}{0}$

$\lim_{x\to 1} \cdots$ DNE

assume $x = 1.0001$, $x^2 - 2x + 1 > 0$

$\lim_{x\to 1^+} \dfrac{x^3+3x+7}{x^2-2x+1} = \infty$

assume $x = 0.99999$, $x^2 - 2x + 1 > 0$

$\lim_{x\to 1^-} \dfrac{x^3+3x+7}{x^2-2x+1} = \infty \Rightarrow \lim_{x\to 1} \dfrac{x^3+3x+7}{x^2-2x+1} = \infty$

Removable continuity

- **Theorem**

> Suppose $f(x) = g(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, and $\lim_{x\to a} g(x)$ exists, then $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$.

这是一些化简的原理，化简后减去了 $x = a$ 不存在的那个点

$$f(x) = \frac{x^2 - 1}{x - 1}, \quad g(x) = x + 1$$

我们观察：

- 当 $x \neq 1$，可以约分：

$$f(x) = \frac{(x-1)(x+1)}{x-1} = x + 1 = g(x)$$

- 但在 $x = 1$ 处，$f(x)$ 是未定义的，$g(x)$ 是定义好的 $g(1) = 2$

现在：

- 在 $x = 1$ 附近，$f(x) = g(x)$
- $\lim_{x\to 1} g(x) = 2 \Rightarrow \lim_{x\to 1} f(x) = 2$

这就是这个定理的应用！

当分子分母都为0的时候，不为从 conclude limit 是 DNE

Example 2.18. Find $\lim_{x\to 1} \dfrac{x^2-1}{x-1}$. $\frac{0}{0}$

$= \lim_{x\to 1} \frac{(x+1)(x-1)}{x-1}$

$= \lim_{x\to 1} (x+1)$

$= 2$

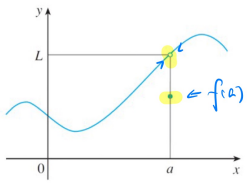
Example 2.19. Find $\lim_{x\to 2} \dfrac{x^2+3x-10}{x^2+x-6}$.

$= \lim_{x\to 2} \frac{(x+5)(x-2)}{(x+3)(x-2)}$

$= \lim_{x\to 2} \frac{(x+5)}{(x+3)}$

$= \frac{7}{5} = 1.4$

Example 2.20. Given $g(x) = \begin{cases} \dfrac{x-2}{x^2-4}, & x \neq 2 \\ 6, & x = 2 \end{cases}$, find $\lim_{x\to 2} g(x)$.



(b) $\lim_{x\to a} f(x) = L$

$g(2) = 6$. function value 和 limit value 不一定一样

$f(a) \neq L$

$\lim_{x\to 2} \frac{x-2}{x^2-4} = \lim_{x\to 2} \frac{x-2}{(x+2)(x-2)}$

$= \lim_{x\to 2} \frac{1}{x+2}$

$= \frac{1}{4}$

Example 2.21. Find $\lim_{x\to 1} \dfrac{x-1}{\sqrt{x}-1}$.

Method 1 : factorization

$= \lim_{x\to 1} \frac{(\sqrt{x}-1)(\sqrt{x}+1)}{\sqrt{x}-1}$

$= \lim_{x\to 1} \sqrt{x}+1$

$= 2$

Method 2 : Rationalization

$= \lim_{x\to 1} \frac{(x-1)(\sqrt{x}+1)}{(\sqrt{x}-1)(\sqrt{x}+1)}$

$= \lim_{x\to 1} \frac{(x-1)(\sqrt{x}+1)}{x-1}$

remove the square root

$= \lim_{x\to 1} \sqrt{x}+1$

$= 2$

Homework

Example 2.22. Evaluate $\lim_{x\to 0}\dfrac{\sqrt{x^2+9}-3}{x^2}$.

rationalization

$$= \lim_{x\to 0}\frac{(\sqrt{x^2+9}-3)(\sqrt{x^2+9}+3)}{x^2(\sqrt{x^2+9}+3)}$$

$$= \lim_{x\to 0}\frac{x^2+9-9}{x^2(\sqrt{x^2+9}+3)}$$

$$= \lim_{x\to 0}\frac{1}{\sqrt{x^2+9}+3} = \frac{1}{6}$$

- Some limits are best calculated by first finding the **left-** and **right-**hand limits.

Example 2.23. Find the limit $\lim_{x\to 0}|x|$.

$$|x| = \begin{cases} -x, & \text{if } x<0 \\ x, & \text{if } x\geq 0 \end{cases}$$

$$\lim_{x\to 0^-}|x| = \lim_{x\to 0^-}(-x) = 0$$

$$\lim_{x\to 0^+}|x| = \lim_{x\to 0^+}(x) = 0$$

$\Rightarrow \lim_{x\to 0}|x| = 0$ since $\lim_{x\to 0^-}|x| = \lim_{x\to 0^+}|x| = 0$

Example 2.24. Does the limit $\lim_{x\to 0}\dfrac{|x|}{x}$ exists?

$$\lim_{x\to 0^-}\frac{|x|}{x} = \lim_{x\to 0^-}\frac{-x}{x} = -1$$

$$\lim_{x\to 0^+}\frac{|x|}{x} = \lim_{x\to 0^+}\frac{x}{x} = 1$$

$\Rightarrow \lim_{x\to 0}\frac{|x|}{x}$ DNE since $\lim_{x\to 0^-}\frac{|x|}{x} \neq \lim_{x\to 0^+}\frac{|x|}{x}$

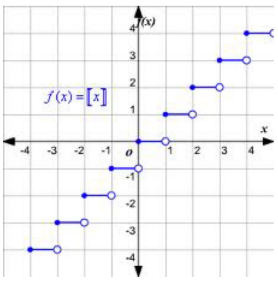
Example 2.25. Given $f(x) = \begin{cases} \sqrt{x-2}, & x>6 \\ x-4, & x<6 \end{cases}$, find $\lim_{x\to 6} f(x)$.

$$\lim_{x\to 6^-} f(x) = \lim_{x\to 6^-}(x-4) = 2$$

$$\lim_{x\to 6^+} f(x) = \lim_{x\to 6^+}(\sqrt{x-2}) = 2$$

$\lim_{x\to 6} f(x) = 2$ since $\lim_{x\to 6^-} f(x) = \lim_{x\to 6^+} f(x)$

Example 2.26. Find $\lim_{x\to -2} [\![x]\!]$.



$\lim_{x\to -2^-} [\![x]\!] = -3$

$\lim_{x\to -2^+} [\![x]\!] = -2$

$\lim_{x\to -2} [\![x]\!]$ DNE. since $\lim_{x\to -2^-} [\![x]\!] \neq \lim_{x\to -2^+} [\![x]\!]$

Suppose $f(x) \leq g(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, then $\lim_{x\to a} f(x) \leq \lim_{x\to a} g(x)$.

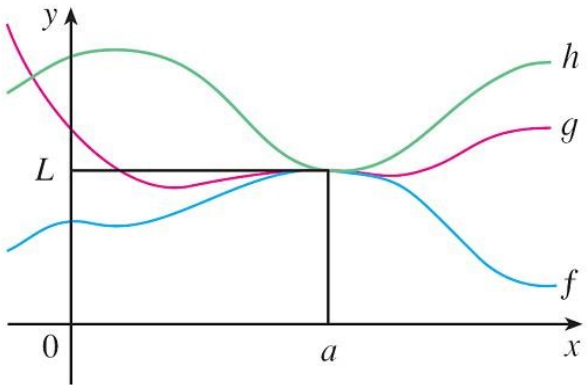
- **Squeeze Theorem/Sandwich Theorem** 一般是用于证明极限

Suppose $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, and

$$\lim_{x\to a} f(x) = \lim_{x\to a} h(x) = L$$

then

$$\lim_{x\to a} g(x) = L$$



Example 2.27. Find $\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right)$.

$-1 \le \sin(\frac{1}{x}) \le 1$ for $x \neq 0$

$-x^2 \le x^2 \sin(\frac{1}{x}) \le x^2$ since $x^2 \geq 0$

$\lim_{x\to 0} -x^2 = 0$

$\lim_{x\to 0} x^2 = 0$

By the squeeze theorem, $\lim_{x\to 0} x^2 \sin(\frac{1}{x}) = 0$

try it !!!

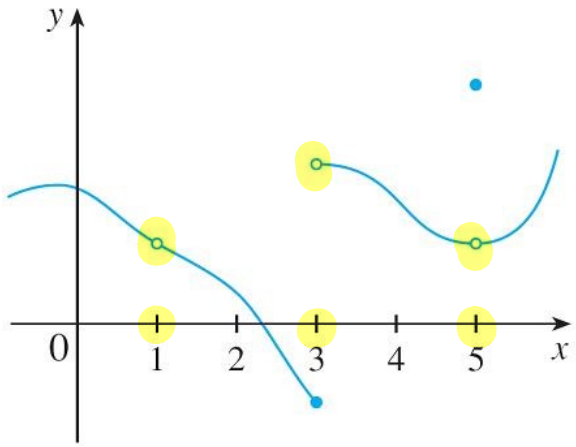
Exercises 2.3 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, 27, 29, 31, 37, 39, 41, 43, 47, 51, 53, 59, 61 (*Hint*: $0 \le f(x) \le x^2$ for all x.).

2.5 Continuity of Functions

- **Definition**

We say that f is continuous at a number a if

$$\lim_{x \to a} f(x) = f(a)$$



} discontinuous point

let the function as $f(x)$. $f(x)$ is discontinuous at

1) $x=1$, since $f(1)$ is not defined.

2) $x=3$, since $\lim_{x \to 3^-} f(x) \neq \lim_{x \to 3^+} f(x)$, $\lim_{x \to 3} f(x)$ DNE

3) $x=5$, since $\lim_{x \to 5} f(x) \neq f(5)$

Example 2.28. Determine whether the function $f(x) = \dfrac{1}{x-2}$ is continuous at $x = 2$.

$f(x)$ is discontinuous at $x=2$ since $f(2)$ is undefined.

$\lim_{x\to 2} \frac{1}{x-2}$ DNE

Example 2.29. Determine whether the function $f(x) = \dfrac{x^2-4}{x-2}$ is continuous at $x = 2$.

$f(x)$ is discontinuous at $x=2$ since $f(2)$ is undefined.

$\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$

example 2.4 用 one side limit ∵ 兩邊極不存的疑惑

Example 2.30. Determine whether the function $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ 2, & x = 2 \end{cases}$ is continuous at $x = 2$.

$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x+2)(x-2)}{x-2} = \lim_{x\to 2} x+2 = 4$

跳走點 極限存在 一側的 ⇒ 不用 one side limit

$f(x)$ is discontinuous at $x=2$, since $\lim_{x\to 2} f(x) \neq f(2)$

Example 2.31. Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ c, & x = 2 \end{cases}$. Find the value of c such that f is continuous at $x = 2$ (*removing the discontinuity*).

$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2} \frac{(x+2)(x-2)}{(x-2)} = \lim_{x\to 2}(x+2) = 4$

To let $f(x)$ continuous at $x=2$, $\lim_{x\to 2} \frac{x^2-4}{x-2} = f(2)$, such that

$\lim_{x\to 2} \frac{x^2-4}{x-2} = c$, $c = 4$

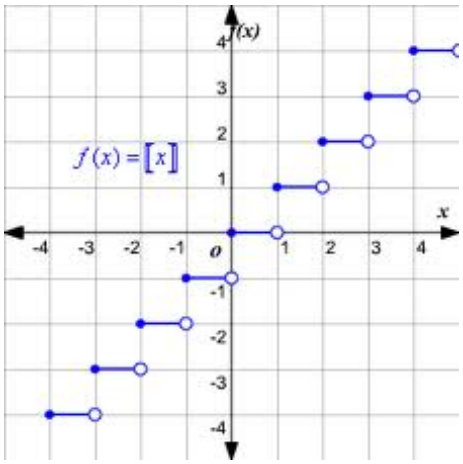
Homework

Example 2.32. Determine all points of discontinuity of $f(x) = \dfrac{\sin x}{x(1-x)}$.

always continuous

$f(x)$ is discontinuous when $x = 0$ and $x = 1$ ($y = x(1-x) = 0$)

Example 2.33. Determine all points of discontinuity of $f(x) = [\![x]\!]$.



$f(x)$ is discontinuous when $x \in \mathbb{Z}$,

Since $\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$, when $a \in \mathbb{Z}$

- **Definition**

We say that f is **left continuous** at a number a if

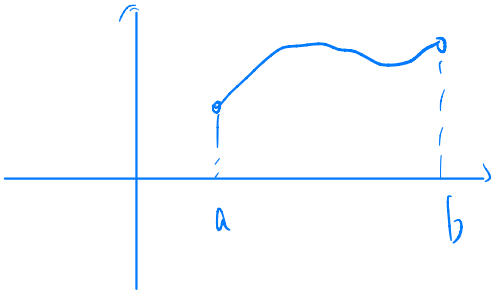
$$\lim_{x \to a^-} f(x) = f(a)$$

We say that f is **right continuous** at a number a if

$$\lim_{x \to a^+} f(x) = f(a)$$

- **Continuity on an Interval**

We say that f is continuous on an open interval if it is continuous at each point of that interval. It is continuous on the closed interval $[a, b]$ if it is continuous on (a, b), right continuous at a and left continuous at b.



- **Theorem**

If f and g are continuous at a, and k is a constant, then the following functions are also continuous at a:

- kf
- $f + g$
- $f - g$
- $f \cdot g$
- $\frac{f}{g}$, provided that $g(a) \neq 0$
- f^n
- $\sqrt[n]{f}$, provided that $f(a) > 0$.

- **Examples of Continuous Functions**

7 Theorem The following types of functions are continuous at every number in their domains:

- polynomials
- rational functions
- root functions
- trigonometric functions
- inverse trigonometric functions
- exponential functions
- logarithmic functions

Example 2.34. Use continuity to evaluate the limit.

(a) $\lim_{x \to \pi} \sin x$ $= \sin \pi = 0$ → Direct substitution

(b) $\lim_{x \to 4} 3^x$ $= 3^4 = 81$

- **Composite Limit Theorem**

composition function : a function inside a function

eg. $f(x) = x^2$; $g(x) = \sin x$

$f \circ g(x) = (\sin x)^2$ (outer function f, inner function g)

$g \circ f(x) = \sin x^2$

eg $\cos\left(\lim_{x\to a}(\quad)\right)$ $\sin\left(\sin\frac{1}{\quad}\right)$

If $\lim_{x\to a} g(x) = L$ and f is continuous at L, then

$$\lim_{x\to a} f\left(g(x)\right) = f\left(\lim_{x\to a} g(x)\right) = f(L)$$

In particular, if g is continuous at a and f is continuous at $g(a)$, then the composite $f \circ g$ is continuous at a.

Example 2.35. Show that $h(x) = \left|x^2 - 3x - 6\right|$ is continuous at each real number.

let $h(x) = g \circ f(x)$, then $g(x) = |x|$, $f(x) = x^2 - 3x - 6$

both f and g have the domain $x \in \mathbb{R}$, Thus, they are both continuous on $\mathbb{R}$

Thus, $h(x) = g \circ f(x)$ is continuous on $\mathbb{R}$

Extra : $h(x) = \sin\left(\frac{3}{x}\right)$ continuous $\forall x \in \mathbb{R} \setminus \{0\}$

Extra : $h(x) = \frac{3}{\sin x}$

$f(x) = \sin x$

$g(x) = \frac{3}{x}$

$g \circ f(x) = h(x)$

$\sin x \neq 0$

$x \neq n\pi$ when $n \in \mathbb{Z}$

Example 2.36. Show that $h(x) = \sin \dfrac{x^4 - 3x + 1}{x^2 - x - 6}$ is continuous except at 3 and -2.

$h(x) = g \circ f(x)$

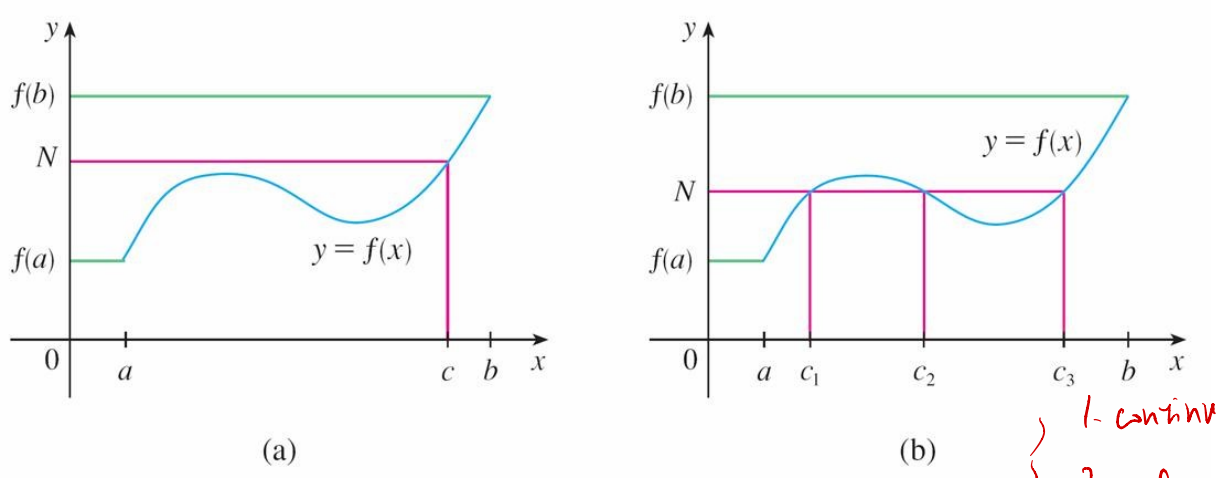
$g(x) = \sin x$

$f(x) = \dfrac{x^4 - 3x + 1}{x^2 - x - 6}$ is undefined when $x = 3$ and $x = -2$ and

this cause discontinuous at this 2 points

- **Intermediate Value Theorem**

> Let f be a function defined on $[a, b]$ and let N be a number between $f(a)$ and $f(b)$. If f is continuous on $[a, b]$, then there is at least a number c between a and b such that $f(c) = N$.



1. continuity
2. f(x)
3. Conclusion

Example 2.37. Show that the equation $x - \cos x = 0$ has a solution between $x = 0$ and $x = \frac{\pi}{2}$.

$f(x)$ N

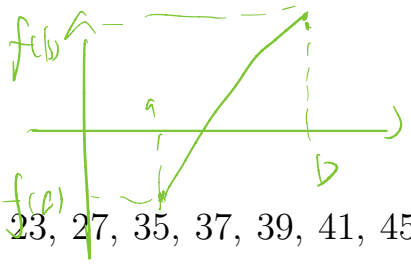
let $f(x) = x - \cos x$, which is continuous on $x \in [0, \frac{\pi}{2}]$, since x and $\cos x$ are continuous on $\mathbb{R}$

$f(0) = -1$; $f(\frac{\pi}{2}) = \frac{\pi}{2}$ ∈ 用 calculator 算出來的時候要注意的技術 looking 的技術

這要注意嗎?

By the IVT, there exists of at lease one number $X = C$, where $0 < c < \frac{\pi}{2}$

such that $f(c) = 0$



Exercises 2.5 3, 11, 13, 15, 17, 19, 21, 23, 27, 35, 37, 39, 41, 45, 47, 51, 53, 55, 69 (*Hint*: $x^3 + 1 = x$).

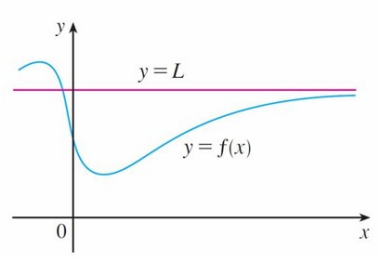
Try it !!!

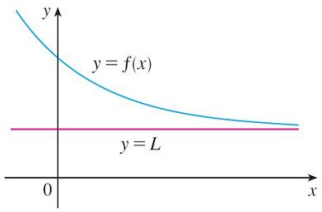
2.6 Limits at Infinity, Horizontal Asymptotes

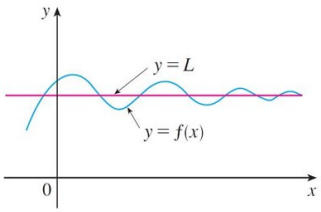
- **Definition** Rationalization

$$\lim_{x\to\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

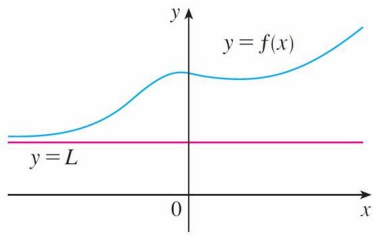


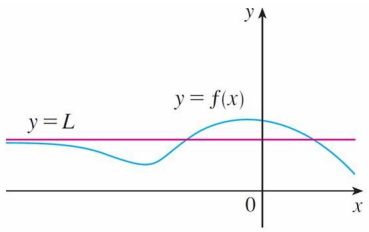




$$\lim_{x\to-\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.





Example 2.38. Find $\lim_{x\to\infty} \dfrac{x}{1+x^2}$.

$$\lim_{x\to\infty} \frac{x\cdot\frac{1}{x}}{x^2\left(\frac{1}{x^2}+1\right)} = \lim_{x\to\infty} \frac{\frac{1}{x}}{\frac{1}{x^2}+1} = \frac{0}{0+1} = 0$$

분자/분모 rational 하게 하기

have a non-zero number in denominator

Example 2.39. Find $\lim_{x\to\infty} \dfrac{2x^3}{1+x^3}$.

$= \lim_{x\to\infty} \dfrac{2x^3}{x^3\left(\frac{1}{x^3}+1\right)}$

$= 2$

Example 2.40. Find $\lim_{n\to\infty} \sqrt{\dfrac{n+1}{n+2}}$.

$= \lim_{n\to\infty} \dfrac{\sqrt{(n+1)\frac{1}{n}\cdot n}}{\sqrt{(n+2)\frac{1}{n}\cdot n}}$

$= \lim_{n\to\infty} \dfrac{\sqrt{1+\frac{1}{n}}\sqrt{n}}{\sqrt{1+\frac{2}{n}}\sqrt{n}}$

$= 1$

Example 2.41. Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+2x}-x\right)$. $\infty - \infty$ Another Indeterminate form

=> multiply with a conjugate.

$= \lim_{x\to\infty} \left(\sqrt{x^2}\sqrt{\frac{x^2}{x^2}+\frac{2x}{x^2}}-x\right)$

$= \lim_{x\to\infty} \left(x\sqrt{1+\frac{2}{x}}-x\right)$

$= \lim_{x\to\infty} x\left(\sqrt{1+\frac{2}{x}}-1\right)$

constant 0

$= 0\,(\infty)$

it's another Indeterminate form with $0\times\infty$

Example 2.39. Find $\lim_{x\to\infty} \dfrac{2x^3}{1+x^3}$.

Example 2.40. Find $\lim_{n\to\infty} \sqrt{\dfrac{n+1}{n+2}}$.

Example 2.41. Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+2x} - x\right)$.

Method 2: rationalization

$$= \lim_{x\to\infty} \left(\sqrt{x^2+2x} - x\right) \times \frac{\sqrt{x^2+2x}+x}{\sqrt{x^2+2x}+x}$$

transform into rational form

$$= \lim_{x\to\infty} \frac{x^2+2x-x^2}{\sqrt{x^2+2x}+x}$$

$$= \lim_{x\to\infty} \frac{2x}{x\sqrt{1-\frac{1}{x}}+x}$$

$$= \lim_{x\to\infty} \frac{2}{\sqrt{1-\frac{1}{x}}+1}$$

$$= 1$$

$$\lim_{x\to-\infty}\frac{\sqrt{x^2+1}}{x+1} = \lim_{x\to-\infty}\frac{\sqrt{x^2}\sqrt{1+\frac{1}{x^2}}}{x\left(1+\frac{1}{x}\right)}$$

∞

$$\sqrt{x^2} = |x| = \begin{cases} -x, & x \le 0 \\ x, & x > 0 \end{cases}$$

$$= \lim_{x\to-\infty}\frac{-x\sqrt{1+\frac{1}{x^2}}}{x\left(1+\frac{1}{x}\right)}$$

∞

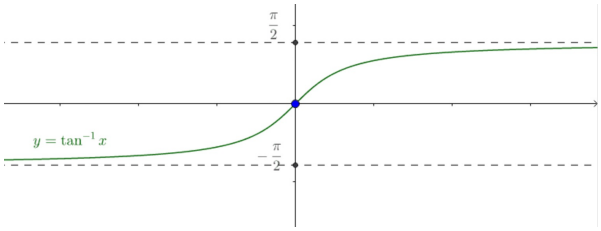
$$= -1$$

know the nature of these functions

Example 2.42. Evaluate $\lim_{x\to 2^+} \arctan\left(\frac{1}{x-2}\right)$.

$= \lim_{x\to 2^+} \arctan(\infty)$

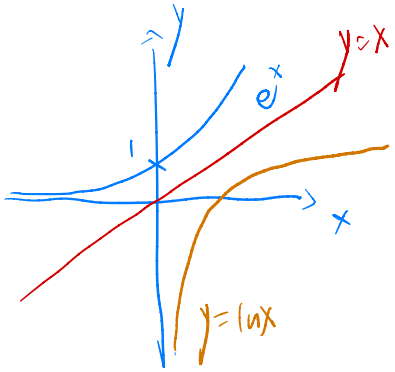
$= \frac{\pi}{2}$



Example 2.43. Find $\lim_{x\to\infty} e^x$ and $\lim_{x\to-\infty} e^x$.

$\lim_{x\to\infty} e^x = \infty$

$\lim_{x\to-\infty} e^x = 0$

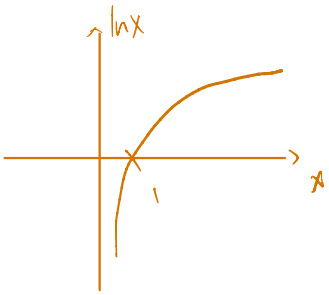


Example 2.44. Find $\lim_{x\to\infty} \ln x$ and $\lim_{x\to 0^+} \ln x$.

$\ln x = \log_e x$

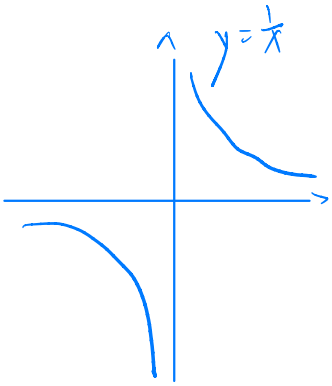
$\lim_{x\to\infty} \ln x = \infty$

$\lim_{x\to 0^+} \ln x = -\infty$



Example 2.45. Find $\lim_{x\to 0^-} e^{1/x}$

$\lim_{x\to 0^-} e^{\frac{1}{x}} = \lim_{\theta\to-\infty} e^{\theta} = 0$



- **Horizontal Asymptotes**

The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x\to\infty} f(x) = L \qquad \text{or} \qquad \lim_{x\to-\infty} f(x) = L$$

Example 2.46. Find the horizontal asymptote of the curve $y = \dfrac{x^2-2}{2x^2+x}$.

$$\lim_{x\to\infty} \frac{x^2-2}{2x^2+x} = \lim_{x\to\infty} \frac{x^2\left(1-\frac{2}{x^2}\right)}{x^2\left(2+\frac{1}{x}\right)} = \lim_{x\to\infty} \frac{1-\frac{2}{x^2}}{2+\frac{1}{x}} = \frac{1}{2}$$

$$\lim_{x\to-\infty} \frac{x^2-2}{2x^2+x} = \lim_{x\to-\infty} \frac{1-\frac{2}{x^2}}{2+\frac{1}{x}} = \frac{1}{2}$$

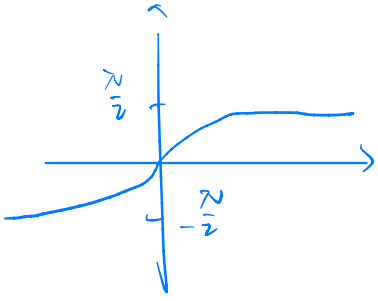
The curve has a horizontal asymptote line at $y = \frac{1}{2}$

Example 2.47. Find the horizontal asymptotes of the curve $y = \tan^{-1} x$.

$$\lim_{x\to\infty} \tan^{-1} x = \frac{\pi}{2}$$

$$\lim_{x\to-\infty} \tan^{-1} x = -\frac{\pi}{2}$$

$\Rightarrow$ conclude from graph (nature)



∴ The curve has 2 horizontal asymptote lines at $y = \frac{\pi}{2}$ and $y = -\frac{\pi}{2}$

Example 2.48. Find the horizontal and vertical asymptotes of the graph of the function $f(x) = \dfrac{\sqrt{2x^2+1}}{3x-5}$.

hori:

$$\lim_{x\to\infty} \frac{\sqrt{2x^2+1}}{3x-5} = \lim_{x\to\infty} \frac{|x|\sqrt{2+\frac{1}{x^2}}}{x(3-\frac{5}{x})} = \frac{\sqrt{2}}{3}$$

$$\lim_{x\to-\infty} \frac{\sqrt{2x^2+1}}{3x-1} = -\frac{\sqrt{2}}{3}$$

The curve has 2 horizontal asymptotes lines at

$y = \frac{\sqrt{2}}{3}$ and $y = -\frac{\sqrt{2}}{3}$

For $f(x)$, $3x-5 \neq 0$

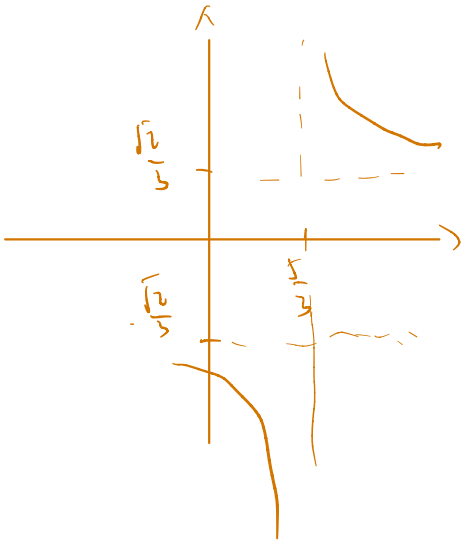
$x \neq \frac{5}{3}$

$$\lim_{x\to\frac{5}{3}} \frac{\sqrt{2x^2+1}}{3x-5} \quad \text{DNE}$$

$$\lim_{x\to\frac{5}{3}^-} \frac{\sqrt{2x^2+1}}{3x-5} = -\infty$$

or $\lim_{x\to\frac{5}{3}^+} \frac{\sqrt{2x^2+1}}{3x-5} = \infty$

$f(x)$ has a vertical asymptotes at $x = \frac{5}{3}$



Exercises 2.6 3, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39 (Hint: Squeeze Theorem), 41, 47, 49, 51.

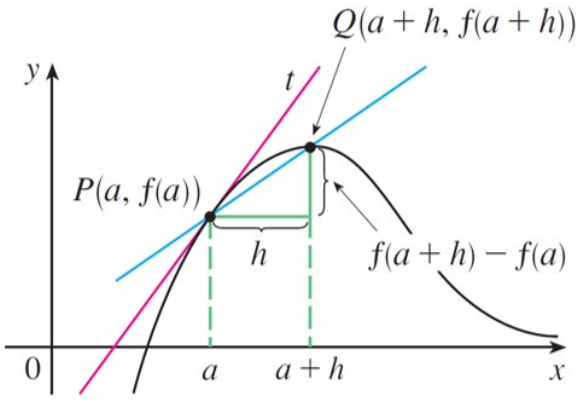
2.7 Derivatives and Rates of Change

- **Tangent Line**

The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is that line through P with slope

$$m_{\tan} = \lim_{h \to 0} m_{\sec} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

provided that this limit exits and is not ∞ or $-\infty$.



- **Derivative**

The derivative of a function f at a number a is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- If the limit exists, we say that f is **differentiable** at a. Finding a derivative is called **differentiation**.

Example 2.49. Given $f(x) = x^3 + 7x$, find $f'(a)$.

Exercises 2.7 5, 7, 14 (6.28 m/s, $(10 - 3.72a)$ m/s, 5.4 s, -10 m/s), 21, 31, 33, 35, 37, 39, 41, 55, 59, 60 (exist, why?).

2.8 The Derivative as a Function

- **Definition**

> The derivative of a function f is another function f' whose value at any number x is
>
> $$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Example 2.50. Given $f(x) = \dfrac{1}{x}$, find $f'(x)$.

$(x^{-1})' = -x^{-2}$

$\dfrac{1}{-x^2}$

$f(x+h) = \dfrac{1}{x+h}$

$f'(x) = \lim_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

$= \lim_{h \to 0} \dfrac{\frac{1}{x+h} - \frac{1}{x}}{h}$

$= \lim_{h \to 0} \dfrac{\frac{x-x-h}{(x+h)x}}{h} = \lim_{h \to 0} -\dfrac{1}{(x+h)x} = -\dfrac{1}{x^2}$

Example 2.51. Given $f(x) = \sqrt{x}$, find $f'(x)$.

- **Leibniz Notation for the Derivative**

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = f'(x)$$

- **Other Notations** For $y = f(x)$

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x) = Df(x) = D_x f(x)$$

- A function is differentiable on an open interval if it is differentiable at every number in that interval.

看左右极限(导数)相同

Example 2.52. Where is the function $f(x) = |x|$ differentiable?

$$f(x) = |x| = \begin{cases} x, & \text{if } x > 0 \\ -x, & \text{if } x \le 0 \end{cases}$$

$$f'(0) = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h}$$ DNE since left and right limits are not equal

$$\lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = \frac{-h - 0}{h} = -1$$

$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \frac{h - 0}{h} = 1$$

$f(x)$ is differentiable for all $\forall x \in \mathbb{R} \setminus \{0\}$

极限 & 导数

极限存在：

$$\lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = L \Rightarrow f(x) \text{在 } a \text{ 连续}$$

导数存在

$$\lim_{h\to 0^-} \frac{f(a+h)-f(a)}{h} = \lim_{h\to 0^+} \frac{f(a+h)-f(a)}{h} = L$$

↓

$f'(x)$ 存在，$f(x)$ 在 $x=a$ 点 differentiable

- **Differentiability Implies Continuity**

> If $f'(a)$ exists, then f is continuous at a.

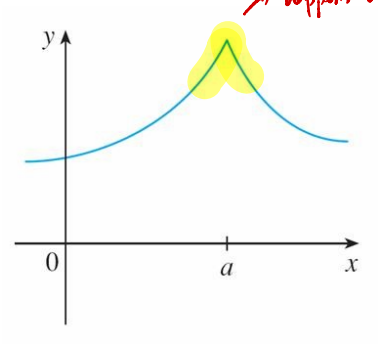
- Continuity does **not** imply differentiability.

$f(x) = |x|$ is continuous all the time but is not differentiable at $x = 0$

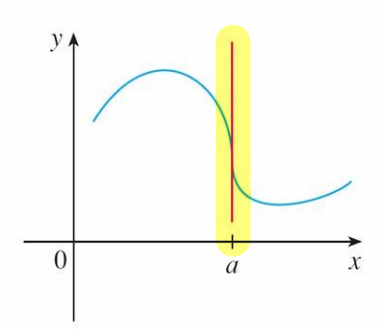
- How can a function fail to be differentiable?

  - A corner

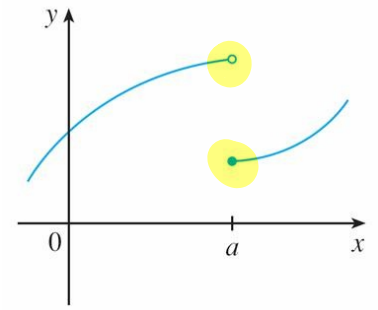
happens only when the formular becomes different



  - A vertical tangent



  - A discontinuity



- **High Order Derivatives**

  The derivative of f is f'. Differentiating f' will produce f'', called the second derivative of f. In the same way, we define f''', the third derivative of f, $f^{(4)}$, the fourth derivative of f, $f^{(5)}$, the fifth derivative of f, and so on.

- **Notations**

There common type.

Derivative	f' Notation	y' Notation	D Notation	Leibniz Notation
First	$f'(x)$	y'	$D_x y$	$\frac{dy}{dx}$
Second	$f''(x)$	y''	$D_x^2 y$	$\frac{d^2y}{dx^2}$
Third	$f'''(x)$	y'''	$D_x^3 y$	$\frac{d^3y}{dx^3}$
Fourth	$f^{(4)}(x)$	$y^{(4)}$	$D_x^4 y$	$\frac{d^4y}{dx^4}$
$\vdots$	$\vdots$	$y^4 \neq y^{(4)}$ $\vdots$	$\vdots$	$\vdots$
nth	$f^{(n)}(x)$	$y^{(n)}$	$D_x^n y$	$\frac{d^n y}{dx^n}$

Exercises 2.8 3, 21, 25, 27, 29, 41, 43, 49, 51, 63.